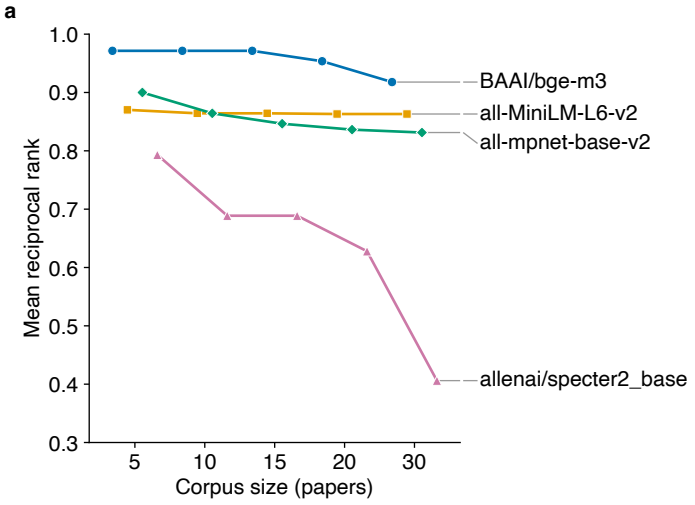
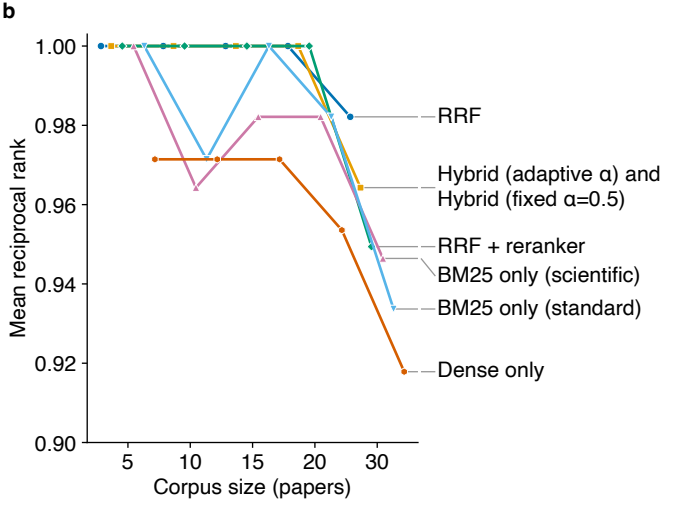
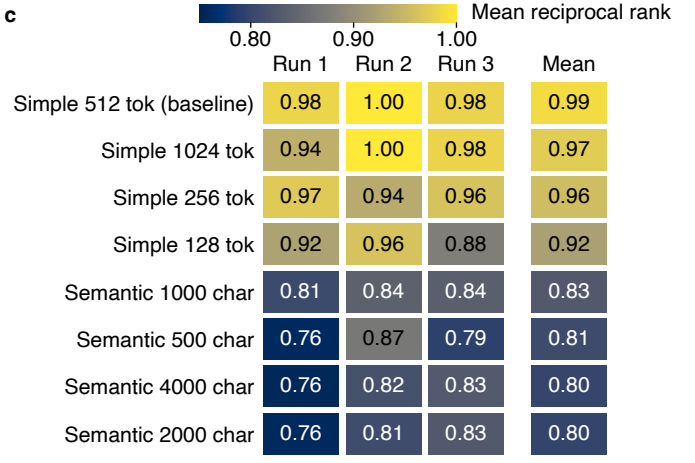
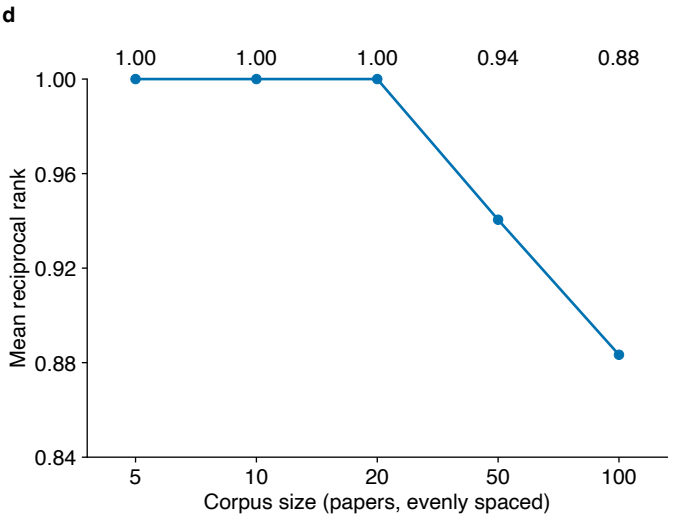

a
Mean reciprocal rank
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
5
10
15
20
30
Corpus size (papers)
BAAI/bge-m3
all-MiniLM-L6-v2
all-mpnet-base-v2
allenai/specter2_base


b
Mean reciprocal rank
1.00
0.98
0.96
0.94
0.92
0.90
5
10
15
20
30
Corpus size (papers)
RRF
Hybrid (adaptive α) and
Hybrid (fixed α=0.5)
RRF + reranker
BM25 only (scientific)
BM25 only (standard)
Dense only


c
Mean reciprocal rank
0.80
0.90
1.00
Run 1
Run 2
Run 3
Mean
Simple 512 tok (baseline) 0.98 1.00 0.98 0.99
Simple 1024 tok 0.94 1.00 0.98 0.97
Simple 256 tok 0.97 0.94 0.96 0.96
Simple 128 tok 0.92 0.96 0.88 0.92
Semantic 1000 char 0.81 0.84 0.84 0.83
Semantic 500 char 0.76 0.87 0.79 0.81
Semantic 4000 char 0.76 0.82 0.83 0.80
Semantic 2000 char 0.76 0.81 0.83 0.80


d
1.00
1.00
1.00
0.94
0.88
Mean reciprocal rank
1.00
0.96
0.92
0.88
0.84
5
10
20
50
100
Corpus size (papers, evenly spaced)